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AUTOMATED ALIGNMENT OF REFUSE COLLECTION VEHICLE

Abstract

A system can include one or more processing circuits to detect a presence of a refuse can, determine a position of the refuse can with respect to a refuse collection vehicle, generate a trajectory that includes at least one path for the refuse collection vehicle to traverse to align the refuse collection vehicle with the refuse can, and present a user interface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/539,658, filed Dec. 14, 2023, which is a continuation application of U.S. patent application Ser. No. 17/232,909, filed Apr. 16, 2021, which claims the benefit of U.S. Provisional Patent Application No. 63/011,616, filed Apr. 17, 2020, the entire contents of each of which are hereby incorporated by reference herein.

BACKGROUND

[0002] Refuse vehicles collect a wide variety of waste, trash, and other material from residences and businesses. Operators of the refuse vehicles transport the material from various waste receptacles within a municipality to a storage or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.).

SUMMARY

[0003] One implementation of the present disclosure is a system for detecting and engaging a refuse can. The system includes at least one sensor positioned on a refuse collection vehicle and configured to detect objects on one or more sides of the refuse vehicle, an actuator assembly coupled to the refuse collection vehicle and configured to actuate to engage the refuse can, and a controller configured to detect, using a single-stage object detector, the presence of the refuse can based on first data received from the at least one sensor, determine, based on the first data, a position of the refuse can with respect to the refuse collection vehicle, generate a first trajectory from the refuse collection vehicle to the position of the refuse can, generate a second trajectory for the actuator assembly, the second trajectory indicating a series of movements to be executed by the actuator assembly to engage the refuse can, and initiate a control action to move at least one of the refuse collection vehicle along the first trajectory or the actuator assembly along the second trajectory to engage the refuse can.

[0004] Another implementation of the present disclosure is a method for detecting a refuse can. The method includes receiving data from one or more sensors positioned on a refuse collection vehicle, processing the data by via a single-stage object detector to identify the refuse can, determining a position of the refuse can with respect to the refuse collection vehicle, generating a first trajectory from the refuse collection vehicle to the position of the refuse can, generating a second trajectory for the actuator assembly, the second trajectory indicating a series of movements to be executed by the actuator assembly to engage the refuse can, and initiating a control action to move at least one of the refuse collection vehicle along the first trajectory or the actuator assembly along the second trajectory to engage the refuse can.

[0005] Yet another implementation of the present disclosure is a controller for a refuse collection

vehicle. The controller includes one or more memory devices having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to perform operations including detecting, using a single-stage object detector, the presence of the refuse can based on first data received from at least one sensor positioned on an exterior of the refuse collection vehicle, determining, based on the first data, a position of the refuse can with respect to the refuse collection vehicle, generating a first trajectory from the refuse collection vehicle to the position of the refuse can, generating a second trajectory for the actuator assembly, the second trajectory indicating a series of movements to be executed by the actuator assembly to engage the refuse can, and presenting, via a screen positioned in a cab of the refuse collection vehicle, a graphical user interface indicating at least one of the first trajectory or the second trajectory.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

[0007] FIGS. 1A and 1B are perspective views of a refuse vehicle, according to some embodiments.

[0008] FIG. 2A is a perspective view of a first type of actuator assembly for use with the refuse vehicle of FIGS. 1A and 1B, according to some embodiments.

[0009] FIG. 2B is a perspective view of a second type of actuator assembly for use with the refuse vehicle of FIGS. 1A and 1B, according to some embodiments.

[0010] FIGS. 3A-3C are examples configurations of the refuse vehicle of FIGS. 1A and 1B, according to some embodiments.

[0011] FIG. 4 a block diagram of a controller for use with a refuse vehicle, according to some embodiments.

[0012] FIGS. 5 and 6 are example architectures for a model implemented by the controller of FIG. 5 for detecting refuse cans, according to some embodiments.

[0013] FIG. 7 is a process for detecting refuse cans, according to some embodiments.

[0014] FIGS. 8A and 8B are example interfaces illustrating the detection of multiple refuse cans, according to some embodiments.

[0015] FIG. 9 is process for initiating control actions based on a detected refuse can, according to some embodiments.

[0016] FIG. 10A is an example diagram of a method for determining a position of a refuse can, according to some embodiments.

[0017] FIG. 10B is an example diagram for determining a trajectory of the refuse vehicle of FIGS. 1A and 1B based on a position of the refuse can, according to some embodiments.

[0018] FIGS. 11A-11D are example user interfaces for presenting a determined trajectory for the refuse vehicle, according to some embodiments.

DETAILED DESCRIPTION

[0019] The following description includes the best mode presently contemplated for practicing the described implementations. This description is not to be taken in a limiting sense, but rather is made merely for the purpose of describing the general principles of the implementations. The scope of the described implementations should be ascertained with reference to the issued claims.

[0020] Referring generally to the FIGURES, systems and methods for detecting a refuse can are shown, according to various embodiments. The refuse can detection systems may include a

controller configured to receive and process data from a plurality of cameras and/or sensors coupled to a refuse vehicle. The refuse vehicle may be a garbage truck, a waste collection truck, a sanitation truck, etc., configured for side-loading, front loading, or rear loading. The plurality of cameras and/or sensors (e.g., LIDAR, radar, etc.) and the controller may be disposed in any suitable location on the refuse vehicle. The controller may process data from the cameras and/or sensors to detect the presence of refuse cans and/or human beings (e.g., or other objects), for example. The location of an identified refuse may be determined and used to navigate the refuse vehicle and/or an actuator assembly (e.g., an actuator assembly) of the refuse vehicle to engage the refuse can. As denoted herein, a refuse can may include any type of residential, commercial, or industrial refuse can.

[0021] Referring now to FIGS. **1A** and **1B**, a refuse vehicle **10** is shown, according to some embodiments. Refuse vehicle **10** may be a garbage truck, a waste collection truck, a sanitation truck, etc., and may be configured as a side-loading refuse vehicle (e.g., as shown in FIG. **1A**), front loading refuse vehicle (e.g., as shown in FIG. **1B**), or a rear-loading refuse vehicle. In other embodiments, refuse vehicle **10** is another type of vehicle (e.g., a skid-loader, a telehandler, a plow truck, a boom lift, etc.). As shown, refuse vehicle **10** includes a chassis, shown as frame **12**; a body assembly, shown as body **14**, coupled to the frame **12** (e.g., at a rear end thereof, etc.); and a cab, shown as cab **16**, coupled to the frame **12** (e.g., at a front end thereof, etc.). The cab **16** may include various components to facilitate operation of the refuse vehicle **10** by an operator, such as a seat, a steering wheel, hydraulic controls, a graphical user interface (e.g., a touchscreen user interface), switches, buttons, dials, etc.

[0022] As shown, refuse vehicle **10** includes a prime mover, shown as engine **18**, coupled to the frame **12** at a position beneath the cab **16**. Engine **18** is configured to provide power to a series of tractive elements, shown as wheels **20**, and/or to other systems of refuse vehicle **10** (e.g., a pneumatic system, a hydraulic system, etc.). Engine **18** may be configured to utilize one or more of a variety of fuels (e.g., gasoline, diesel, bio-diesel, ethanol, natural gas, etc.), according to various exemplary embodiments. According to an alternative embodiment, engine **18** additionally or alternatively includes one or more electric motors coupled to frame **12** (e.g., a hybrid refuse vehicle, an electric refuse vehicle, etc.). The electric motors may consume electrical power from an on-board storage device (e.g., batteries, ultracapacitors, etc.), from an on-board generator (e.g., an internal combustion engine, etc.), and/or from an external power source (e.g., overhead power lines, etc.) and provide power to the systems of refuse vehicle **10**.

[0023] In some embodiments, refuse vehicle **10** is configured to transport refuse from various waste receptacles within a municipality to a storage and/or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). As shown, the body **14** includes a plurality of panels, shown as panels **32**, a tailgate **34**, and a cover **36**. In some embodiments, as shown in FIG. **1B**, body **14** further includes a door, shown as top door **38**, which is movably coupled along cover **36** to seal the opening thereby preventing refuse from escaping the refuse compartment **30** (e.g., due to wind, bumps in the road, etc.). Panels **32**, tailgate **34**, cover **36**, and/or door **38** define a collection chamber (e.g., hopper, etc.), shown as refuse compartment **30**. Loose refuse may be placed into refuse compartment **30** where it may thereafter be compacted. Refuse compartment **30** may provide temporary storage for refuse during transport to a waste disposal site and/or a recycling facility. In some embodiments, at least a portion of body **14** and refuse compartment **30** extend in front of cab **16**. In some embodiments, body **14** and refuse compartment **30** are positioned behind cab **16**.

[0024] In some embodiments, refuse compartment **30** includes a hopper volume and a storage volume. Refuse may be initially loaded into the hopper volume and thereafter compacted into the storage volume. According to an exemplary embodiment, the hopper volume is positioned between the storage volume and cab **16** (i.e., refuse is loaded into a position of refuse compartment **30** behind cab **16** and stored in a position further toward the rear of refuse compartment **30**). In other embodiments, the storage volume is positioned between the hopper volume and cab **16** (e.g., a rear-

loading refuse vehicle, etc.).

[0025] As shown in FIG. 1A, refuse vehicle **10**, when configured as a side-loading refuse vehicle, may include a side-loading lift mechanism/system (i.e., a side-loading lift assembly), shown as lift assembly **100**. Lift assembly **100** includes a grabber assembly, shown as grabber assembly **42**, slidably coupled to a guide, shown as track **20**, and configured to move along an entire length of the track **20**. Track **20** is shown to extend along substantially an entire height of the body **14** and is configured to cause the grabber assembly **42** to tilt or rotate near an upper height of the body **14**. In other embodiments, the track **20** extends along substantially an entire height of the body **14** on a rear side of the body **14**.

[0026] Grabber assembly **42** is shown to include a pair of actuators, shown as actuators **44**. Actuators **44** are configured to releasably secure a refuse can to grabber assembly **42**, according to an exemplary embodiment. Actuators **44** are selectively repositionable (e.g., individually, simultaneously, etc.) between an engaged position or state and a disengaged position or state. In the engaged position, actuators **44** are rotated towards one other such that the refuse can may be grasped therebetween. In the disengaged position, actuators **44** rotate outwards (e.g., as shown in FIG. 2A) such that the refuse can is not grasped by actuators **44**. By transitioning between the engaged position and the disengaged position, actuators **44** releasably couple the refuse can to grabber assembly **42**.

[0027] In operation, the refuse vehicle **10** may pull up alongside the refuse can, such that the refuse can is positioned to be grasped by the grabber assembly **42** therein. The grabber assembly **42** may then transition into an engaged state to grasp the refuse can. After the refuse can has been securely grasped, the grabber assembly **42** may be transported along the track **20** (e.g., by an actuator) with the refuse can. When the grabber assembly **42** reaches the end of track **20**, grabber assembly **42** may tilt and empty the contents of the refuse can into the refuse compartment **30**. The tilting is facilitated by the path of track **20**. When the contents of the refuse can have been emptied into refuse compartment **30**, the grabber assembly **42** may descend along track **20** and return the refuse can to the ground. Once the refuse can has been placed on the ground, the grabber assembly **42** may transition into the disengaged state, releasing the refuse can.

[0028] As shown in FIG. 1B, refuse vehicle **10**, when configured as a front loading refuse vehicle, may include a front loading lift mechanism/system (i.e., a front loading lift assembly), shown as lift assembly **200**. Lift assembly **200** includes a pair of arms, shown as lift arms **52**, coupled to the frame **12** and/or the body **14** on either side of the refuse vehicle **10** such that the lift arms **52** extend forward of the cab **16** (e.g., a front loading refuse vehicle, etc.). In other embodiments, the lift assembly **200** extends rearward of the body **14** (e.g., a rear-loading refuse vehicle, etc.). In still other embodiments, the lift assembly **200** extends from a side of the body **14** (e.g., a side-loading refuse vehicle, etc.). The lift arms **52** may be rotatably coupled to frame **12** with a pivot (e.g., a lug, a shaft, etc.). As shown, the lift assembly **200** includes first actuators, shown as lift arm actuators **54** (e.g., hydraulic cylinders, etc.), coupled to the frame **12** and the lift arms **52**. The lift arm actuators **54** are positioned such that extension and retraction thereof rotates the lift arms **52** about an axis extending through the pivot, according to an exemplary embodiment.

[0029] An attachment assembly **210** may be coupled to the lift arms **52** of the lift assembly **200**. As shown, the attachment assembly **210** is configured to engage with a first attachment, shown as container attachment **220**, to selectively and releasably secure the container attachment **220** to the lift assembly **200**. In some embodiments, attachment assembly **210** may be configured to engage with a second attachment, such as a fork attachment, to selectively and releasably secure second attachment to the lift assembly **200**. In various embodiments, attachment assembly **210** may be configured to engage with another type of attachment (e.g., a street sweeper attachment, a snow plow attachment, a snowblower attachment, a towing attachment, a wood chipper attachment, a bucket attachment, a cart tipper attachment, a grabber attachment, etc.).

[0030] As shown in FIG. 1B, the lift arms **52** are rotated by the lift arm actuators **54** to lift the

container attachment **220** or other attachment over the cab **16**. Lift assembly **200** includes second actuators, shown as articulation actuators **56** (e.g., hydraulic cylinders, etc.). In some embodiments, the articulation actuators **56** are positioned to articulate the attachment assembly **210**. Such articulation may assist in tipping refuse out of the container attachment **220** and/or a refuse can (e.g., coupled to the lift assembly **200** by a fork attachment, etc.) and into the hopper volume of the refuse compartment **30** through an opening in the cover **36**. The lift arm actuators **54** may thereafter rotate the lift arms **52** to return the empty container attachment **220** to the ground. In some embodiments, top door **38** is movably coupled along the cover **36** to seal the opening thereby preventing refuse from escaping the refuse compartment **30** (e.g., due to wind, bumps in the road, etc.).

[0031] Referring now to FIGS. **2A** and **2B**, detailed perspective views of lift assemblies for use with refuse vehicle **10** are shown, according to some embodiments. Specifically, FIG. **2A** shows a detailed, perspective view of lift assembly **100**, according to some embodiments. As described briefly above, lift assembly **100** includes track **20** and grabber assembly **42**, which includes a frame, chassis, or connecting member, shown as carriage **26**. The track **20** extends along substantially the entire height of the body **14**, according to the exemplary embodiment shown. The body **14** includes a panel, shown as loading section **22**, that defines a cutout or notch, shown as recess **24**, through which the track **20** passes. The recess **24** facilitates a curved portion of the track **20** extending around the top of the loading section **22** without increasing the overall height of the vehicle **10**. When the grabber assembly **42** moves along the curved portion of the track **20**, the grabber assembly **42** is inverted to empty the refuse can releasably coupled to the grabber assembly **42** into the refuse compartment **30**.

[0032] The carriage **26** is slidably coupled to the track **20**. In operation, the carriage **26** may translate along a portion or all of the length of the track **20**. The carriage **26** is removably coupled (e.g., by removable fasteners) to a body or frame of the grabber assembly **42**, shown as grabber frame **46**. Alternatively, the grabber frame **46** may be fixedly coupled to (e.g., welded to, integrally formed with, etc.) the carriage **26**. The actuators **44** are each pivotally coupled to the grabber frame **46** such that they rotate about a pair of axes **45**. The axes **45** extend substantially parallel to one another and are longitudinally offset from one another. In some embodiments, one or more actuators configured to rotate the actuators **44** between the engaged state and the disengaged state are coupled to the grabber frame **46** and/or the carriage **26**.

[0033] Referring now to FIG. **2B**, a detailed, perspective view of lift assembly **200** is shown, according to some embodiments. As shown, container attachment **220** includes a container, shown as refuse can **202**; an articulating refuse collection arm, shown as collection arm assembly **270**; and an interface, shown as attachment interface **280**. The refuse can **202** has a first wall, shown as front wall **210**; an opposing second wall, shown as rear wall **220** (e.g., positioned between the cab **16** and the front wall **210**, etc.); a first sidewall, shown as first sidewall **230**; an opposing second sidewall, shown as second sidewall **240**; and a bottom surface, shown as bottom **250**. The front wall **210**, the rear wall **220**, the first sidewall **230**, the second sidewall **240**, and the bottom **250** cooperatively define an internal cavity, shown as container refuse compartment **260**. According to an exemplary embodiment, the container refuse compartment **260** is configured to receive refuse from a refuse can (e.g., a residential garbage can, a recycling bin, etc.).

[0034] As shown, the second sidewall **240** of the refuse can **202** defines a cavity, shown as recess **242**. The collection arm assembly **270** is coupled to the refuse can **202** and may be positioned within the recess **242**. In other embodiments, the collection arm assembly **270** is otherwise positioned (e.g., coupled to the rear wall **220**, coupled to the first sidewall **230**, coupled to the front wall **210**, etc.). According to an exemplary embodiment, the collection arm assembly **270** includes an arm, shown as arm **272**; a grabber assembly, shown as grabber **276**, coupled to an end of the arm **272**; and an actuator, shown as actuator **274**. The actuator **274** may be positioned to selectively reorient the arm **272** such that the grabber **276** is extended laterally outward from and retracted

laterally inward toward the refuse can **202** to engage (e.g., pick up, etc.) a refuse can (e.g., a garbage can, a reclining bin, etc.) for emptying refuse into the container refuse compartment **260**. [0035] Referring now to FIGS. **3A-3C**, example configurations of refuse vehicle **10** are shown, according to some embodiments. FIGS. **3A-3C** may illustrate examples of potential configurations of refuse vehicle **10** in addition to the configurations described above with respect to FIGS. **1A-1B** and **2A-2B**. Specifically, FIG. **3A** illustrates a front loading configuration of refuse vehicle **10** with an intermediate storage container. FIG. **3B** illustrates another front loading configuration of refuse vehicle **10** with an intermediate storage container that includes an actuator assembly (e.g., similar to container attachment **220**). FIG. **3C** illustrates a side-loading configuration of refuse vehicle **10** (e.g., an auto side-loader) with a grabber-tipper assembly configured to engage an industrial or commercial refuse can. It will be appreciated that the configurations shown in FIGS. **3A-3C** illustrate example configurations of refuse vehicle **10** and are not intended to be limiting. As described above, refuse vehicle **10** may be configured in any number of front, side, and/or rear-loading configurations, with any type of lift and/or grabber assembly for engaging a commercial or residential refuse can.

[0036] Referring now to FIG. **4**, a controller **400** for refuse vehicle **10** is shown, according to some embodiments. Controller **400** may be configured to receive data from image and/or object sensors (i.e., cameras and sensors) to detect and/or track a plurality of refuse can located on any side of a refuse vehicle (e.g., the front, sides, or rear of refuse vehicle **10**). Controller **400** may be further configured to initiate automated control actions based on the detection of a refuse can. It will be appreciated that controller **400** may be implemented via single controller or may be implemented across multiple controllers or devices.

[0037] Controller **400** may be one of one or more controllers of refuse vehicle **10**, for example. Controller **400** generally receives and processes data from one or more image and/or object sensors disposed at various locations of refuse vehicle **10** to identify refuse cans located on at least the curb side of refuse vehicle **10**. Controller **400** is shown to include a processing circuit **402** including a processor **404** and a memory **406**. In some embodiments, processing circuit **402** is implemented via one or more graphics processing units (GPUs). Processor **404** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components. In some embodiments, processor **404** is implemented as one or more graphics processing units (GPUs).

[0038] Memory **406** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **406** can be or include volatile memory or non-volatile memory. Memory **406** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to an example embodiment, memory **406** is communicably connected to processor **404** via processing circuit **402** and includes computer code for executing (e.g., by processing circuit **402** and/or processor **404**) one or more processes described herein.

[0039] Processing circuit **402** can be communicably connected to a network interface **408** and an input/output (I/O) interface **410**, such that processing circuit **402** and the various components thereof can send and receive data via interfaces **408** and **410**. In some embodiments, controller **400** is communicably coupled with a network **440** via network interface **408**, for transmitting and/or receiving data from/to network connected devices. Network **440** may be any type of network (e.g., intranet, Internet, VPN, a cellular network, a satellite network, etc.) that allows controller **400** to communicate with other remote systems. For example, controller **400** may communicate with a server (i.e., a computer, a cloud server, etc.) to send and receive information regarding operations

of controller **400** and/or refuse vehicle **10**.

[0040] Network interface **408** may include any type of wireless interface (e.g., antennas, transmitters, transceivers, etc.) for conducting data communications with network **440**. In some embodiments, network interface **408** includes a cellular device configured to provide controller **400** with Internet access by connecting controller **400** to a cellular tower via a 2G network, a 3G network, an LTE network, etc. In some embodiments, network interface **408** includes other types of wireless interfaces such as Bluetooth, Wi-Fi, Zigbee, etc.

[0041] In some embodiments, controller **400** may receive over-the-air (OTA) updates or other data from a remote system (e.g., a server, a computer, etc.) via network **440**. The OTA updates may include software and firmware updates for controller **400**, for example. Such OTA updates may improve the robustness and performance on controller **400**. In some embodiments, the OTA updates may be received periodically to keep controller **400** up-to-date.

[0042] In some embodiments, controller **400** is communicably coupled to any number of subsystems and devices of refuse vehicle **10** via I/O interface **410**. I/O interface **410** may include wired or wireless interfaces (e.g., antennas, transmitters, transceivers, wire terminals, etc.) for conducting data communications with subsystems and/or devices of refuse vehicle **10**. In some embodiments, I/O interface **410** may include a Controller Area Network (CAN) bus, a Local Interconnect Network (LIN) bus, a Media Oriented Systems Transport (MOST) bus, an SAE J1850 bus, an Inter-Integrated Circuit (I2C) bus, etc., or any other bus commonly used in the automotive industry. As shown, I/O interface **410** may transmit and/or receive data from a plurality of vehicle subsystems and devices including image/object sensors **430**, a user interface **432**, vehicle systems **434**, and/or an actuator assembly **436**.

[0043] As described herein, image/object sensors **430** may include any type of device that is configured to capture data associated with the detection of objects such as refuse cans. In this regard, image/object sensors **430** may include any type of image and/or object sensors, such as one or more visible light cameras, full-spectrum cameras, LIDAR cameras/sensors, radar sensors, infrared cameras, image sensors (e.g., charged-coupled device (CCD), complementary metal oxide semiconductor (CMOS) sensors, etc.), or any other type of suitable object sensor or imaging device. Data captured by image/object sensors **430** may include, for example, raw image data from one or more cameras (e.g., visible light cameras) and/or data from one or more sensors (e.g., LIDAR, radar, etc.) that may be used to detect objects.

[0044] Generally, image/object sensors **430** may be disposed at any number of locations throughout and/or around refuse vehicle **10** for capturing image and/or object data from any direction with respect to refuse vehicle **10**. For example, image/object sensors **430** may include a plurality of visible light cameras and LIDAR cameras/sensors mounted on the forward and lateral sides of refuse vehicle **10** for capturing data as refuse vehicle **10** moves down a path (e.g., a roadway). In some embodiments, one or more of image/object sensors **430** may be located on an attachment utilized by refuse vehicle **10**, such as container attachment **220** described above.

[0045] User interface **432** may be any electronic device that allows an operator to interact with controller **400**. Examples of user interfaces or devices include, but are not limited to, mobile phones, electronic tablets, laptops, desktop computers, workstations, and other types of electronic devices. In some embodiments, user interface **432** is a control system (i.e., a control panel) configured to display information to an operator of refuse vehicle **10** and/or receive user inputs. In this regard, user interface **432** may include at least a display for presenting information to a user and a user input device for receiving user inputs. In one example, user interface **432** includes a touchscreen display panel located in the cab **16** of refuse vehicle **10** and configured to present an operator with a variety of information regarding the operations of refuse vehicle **10**. User interface **432** may further include a user input device, such as a keyboard, a joystick, buttons, etc.

[0046] Vehicle systems **434** may include any subsystem or device associated with refuse vehicle **10**. Vehicle systems **434** may include, for example, powertrain components (e.g., prime mover **18**),

steering components, a grabber arm, lift assemblies, etc. Vehicle system **434** may also include electronic control modules, control units, and/or sensors associated with any systems, subsystems, and/or devices of refuse vehicle **10**. For example, vehicle system **434** may include an engine control unit (ECU), a transmission control unit (TCU), a Powertrain Control Module (PCM), a Brake Control Module (BCM), a Central Control Module (CCM), a Central Timing Module (CTM), a General Electronic Module (GEM), a Body Control Module (BCM), an actuator or grabber assembly control module, etc. In this manner, any number of vehicle systems and devices may communicate with controller **400** via I/O interface **410**.

[0047] Actuator assembly **436** may include at least the components of a lift assembly for engaging, lifting, and emptying a refuse can. Actuator assembly **436** can include, for example, any of the components of lift assembly **100** and/or lift assembly **200**, described above with respect to FIGS. **1A** and **1B**. For example, actuator assembly **436** may include lift assembly **200**, where a fork attachment is coupled to lift assembly **200** for engaging and lifting front loading refuse cans (e.g., industrial or commercial refuse cans, as shown in FIG. **1B**). Actuator assembly **436** may include a plurality of actuators (e.g., linear actuators, lift actuators, horizontal actuators, etc.) for moving to engage the refuse can. As an example, actuator assembly **436** may be configured to move horizontally, vertically, orthogonally, etc., to refuse vehicle **10** in order to engage a refuse can. In some embodiments, actuator assembly **436** may further include an actuator assembly control module, configured to receive data and/or signals from controller **400** to initiate control actions for a grabber arm or actuator.

[0048] Still referring to FIG. **4**, memory **406** is shown to include an object detector **420**. Object detector **420** may generally receive and process data from image/object sensors **430** to detect objects (e.g., refuse cans). It will be appreciated that, as denoted herein, the data received and processed by object detector **420** may include any type of data as described above with respect to image/object sensors **430**, including video from which images and/or other image data can be extracted. As described above, the data may also include data from one or more sensors (e.g., LIDAR, radar, etc.) that may be utilized to detect an object (e.g., a refuse can) and/or a location or position of the object. As shown, for example, object detector **420** may receive data from image/object sensors **430** via I/O interface **410**.

[0049] Object detector **420** may process the received data to detect target objects, including human beings and/or refuse cans. It will be appreciated, however, that object detector **420** may be configured to detect other objects based on other implementations of controller **400**. In this regard, object detector **420** may provide means for controller **400** to detect and track a plurality of refuse cans on a path being traveled by refuse vehicle **10**.

[0050] Object detector **420** may include a neural network or other similar model for processing received data (e.g., from image/object sensors **430**) to detect target objects. As described herein, object detector **420** is generally a one-stage object detector (e.g., deep learning neural network), or may utilize a one-stage object detection method. Unlike two-stage object detectors (e.g., regional convolution neural network (R-CNN), Fast R-CNN, etc.), object detector **420** may process image data in a single stage and may provide advantages over many two-stage detectors such as increased speed (i.e., decreased computing time).

[0051] In a preferred embodiment, object detector **420** implements the architecture of RetinaNet. Details of RetinaNet, according to one implementation, can be found in *Focal Loss for Dense Object Detection* by Lin et. al., published in February 2018 and incorporated herein by reference in its entirety. In this regard, object detector **420** may also provide improvements over other one-stage object detectors, such as you-only-look-once (YOLO) and single shot detectors (SSDs). For example, object detector **420** may provide increased accuracy when compared to many one-stage object detectors, and even when compared to many two-stage detectors. Additionally, object detector **420** may scale better than many other one- and two-stage object detectors (e.g., SSD). The one-stage object detection methods of RetinaNet, as implemented by object detector **420**, are

described in detail below.

[0052] Referring now to FIGS. 5 and 6, examples of the architecture of object detector 420 are shown, according to some embodiments. Referring first to FIG. 5, the overall architecture of object detector 420 is shown. Object detector 420 generally includes a feature pyramid network (FPN) backbone and two task-specific subnetworks. Details of FPNs, according to one implementation, can be found in *Feature Pyramid Network* by Lin et. al., published in January 2017 and incorporated herein by reference in its entirety. Advantageously, FPN is multiscale, semantically strong at all scales, and quick.

[0053] The FPN is built on top of a residual neural network (ResNet) architecture. Details of ResNet, according to one implementation, can be found in *Deep Residual Learning for Image Recognition* by He et. al., published in December 2015 and incorporated herein by reference in its entirety. As shown in FIG. 6, ResNet utilizes a bottleneck architecture. For each residual function, ResNet uses three layers that are 1×1 , 3×3 , and 1×1 convolutions. The 1×1 layers act to decrease and subsequently increase dimensions of an input, while the 3×3 layer acts as a bottleneck.

[0054] Referring again to FIG. 5, at each FPN level, two fully convolutional networks (FCNs) are attached, including a classification subnet and a box regression subnet. The classification and box regression subnets may be attached in parallel, as shown. The classification subnet predicts a probability of the presence of an object at a particular location, whereas the box regression subnet regresses the offset of each anchor box, described below, to a ground-truth object. The design of the classification subnet and the box regression subnet may be similar, where the two subnets have slightly different final convolutional layers. More specifically, final convolutional layer of the classification subnet may include KA filters, where K is the number of object classes (e.g., types of objects such as various types of refuse cans, people, cars, etc.) and A is the number of anchor boxes, and the box regression subnet may include 4A filters.

[0055] Anchor boxes, as mentioned above, define an area of an input image (e.g., input data) and detect an object from multiple (e.g., K) object classes in the area that the anchor box covers. For each anchor, a focal loss is applied during training of the object detector (e.g., object detector 420). The focal loss is a loss function designed to down-weight easily classified portions of an input image (e.g., the background). In this manner, the focal loss concentrates the network on difficult portions of the input image to increase the accuracy of the trained object detector (e.g., object detector 420), while also reducing the time required to train the object detector. For operations after training, the object detector selects a portion of anchor boxes with a confidence score (i.e., probability for each object class that an anchor box contains the object class) above a threshold value for generating bounding box predictions, as shown in FIG. 8A.

[0056] In some embodiments, object detector 420 is post-processed (e.g., during training) by implementing automated augmentation and/or stochastic regularization to renormalize newer versions of object detector 420 that have been trained using new data. Automated augmentation may include, for example, automatically augmenting image data to produce slightly varied versions of the image data to retrain and improve object detector 420. Said post-processing techniques may improve the performance of object detector 420, for example, by reducing overfitting of object detector 420.

[0057] The model implemented by object detector 420 may be trained by any number of methods. For example, object detector 420 may be trained during manufacture or prior to implementation. In some embodiments, initial training of object detector 420 may be handled by a remote system (e.g., a server or computer), and a trained instance of object detector 420 may be implemented via controller 400. Similarly, object detector 420 may be updated or replaced by receiving updated object model data and/or a new version of object detector 420 via an over-the-air (OTA) update from a remote system via network 440. For example, a new version of object detector 420 may be trained on a remote server system and uploaded (i.e., transmitted) to controller 400 via network 440. In this manner, object detector 420 may be continuously improved to provide improved object

detection.

[0058] Referring again to FIG. 4, memory 406 is shown to further include a user interface (UI manager) 422. UI manager 422 may generate a user interface based on data captured by image/object sensors 430 and/or detected object data from object detector 420. UI manager 422 may present a generated user interface via user interface 432, for example. The user interface may include data captured by image/object sensors 430 (e.g., live, delayed, or previously captured image data) and an indication of any detected objects within the data. As an example, the user interface may present an image of a path (e.g., roadway) that refuse vehicle 10 is traveling on, and may indicate one or more detected refuse cans located along the roadway. An example user interface is described in detail below, with respect to FIG. 8A.

[0059] The user interface generated by UI manager 422 may provide means for a user (e.g., an operator of refuse vehicle 10) to interact with refuse vehicle 10 and/or actuator assembly 436 for semi-autonomous or non-autonomous operations. For example, a user interface that indicates two or more refuse cans may provide means for the user to select a particular one of the refuse cans to act on (e.g., to move to and engage). The user interface may also provide other information regarding the operations of refuse vehicle 10, such as alarms, warnings, and or notifications. In some embodiments, the user interface generated by UI manager 422 may include a notification when a human being is detected within a danger zone. This may alert an operator to an unsafe condition and/or may indicate to the operator why automated refuse can collection cannot be implemented (e.g., until no human beings are located in a danger zone).

[0060] Memory 406 is shown to further include a control module 424. Control module 424 determine and/or implement control actions based on detected objects (e.g., from object detector 420) and/or user inputs (e.g., from user interface 432). In some embodiments, control module 424 may implement any number of automated control actions based on detected objects such as refuse cans and/or human beings. In a first example, control module 424 may implement automated collection of a refuse can, based on detection of the refuse can. In this example, once a refuse can is detected, a location of the refuse can may be determined using any number of known methods. Based on the determined location of the target refuse can, control module 424 may determine a trajectory for refuse vehicle 10 and/or actuator assembly 436 in order to engage the refuse can.

[0061] In some embodiments, control module 424 may control (e.g., by transmitting control signals) vehicle systems 434 and/or actuator assembly 436 to move to and engage the refuse can. For example, control module 424 may transmit control signals to any number controllers associated with vehicle systems 434 (e.g., the ECU, the TCU, an automated steering system, etc.) in order to move refuse vehicle 10 to a desired position near a refuse can. In another example, control module 424 may transmit control signals to a controller associated with actuator assembly 436 in order to move/control actuator assembly 436.

[0062] In some embodiments, when a human being is detected within a danger zone (e.g., within a predefined zone and/or distance of refuse vehicle 10 and/or actuator assembly 436), control module 424 may initiate safety actions. The safety actions may include, for example, preventing refuse vehicle 10 and/or actuator assembly 436 from moving to and/or engaging the refuse can while the human being is detected within the danger zone. In some embodiments, control module 424 may initiate an alert/alarm/notification based on the detection of a human being in a danger zone, and may provide an indication of the alert to UI manager 422 for display via user interface 432.

[0063] Still referring to FIG. 4, memory 406 is shown to further include a feedback module 426. Feedback module 426 may receive data from image/object sensors 430 and/or one or more sensors (e.g., inertia sensors, GPS, etc.) associated with vehicle systems 434 and/or actuator assembly 436 to adjust and/or alter a trajectory (i.e., movement) of refuse vehicle 10 or actuator assembly 436. In some embodiments, feedback module 426 may process data (e.g., from image/object sensors 430 and/or object detector 420) to adjust and/or alter a trajectory (i.e., movement) of refuse vehicle 10 or actuator assembly 436. In some embodiments, feedback module 426 may include a model for

processing feedback data. In some such embodiments, the model may be a recurrent neural network (RNN) or other suitable type of neural network for processing feedback data.

[0064] Referring now to FIG. 7, a process **700** for detecting a refuse can from captured image and/or object data is shown, according to some embodiments. Process **700** may be a process implemented by a controller of a refuse vehicle (e.g., refuse vehicle **10**) for detecting one or more refuse cans from data captured by object sensors disposed at various locations of the refuse vehicle. Process **700** may be implemented by controller **400**, as described above, for example.

[0065] At step **702** data is received from one or more image and/or object sensors (e.g., image/object sensors **430**) disposed at various locations of a refuse vehicle. In some embodiments, data is received from at least a visible light camera and a LIDAR camera or sensor. Received data may include raw data from one or more cameras (e.g., visible light cameras) and/or data from one or more sensors (e.g., LIDAR, radar, etc.), as described above. In various embodiments, the data includes still images, video, or other data that can be used to detect an object or objects. In some embodiments, the received data includes at least raw image data and LIDAR data. As described above with respect to FIG. 4, for example, data may be captured from one or more sides of a refuse vehicle, in order to detect refuse cans and/or other objects on either side of a roadway or path that the refuse vehicle traverses.

[0066] At step **704**, the raw data received from the one or more sensors is preprocessed. It will be appreciated that step **704** may be an optional step in some implementations, where preprocessing is necessary or desired. In other implementations, it may not be necessary or desirable to preprocess the data. Accordingly, in some embodiments, preprocessing of data may be implemented prior to processing the data to detect objects such as refuse cans. In various embodiments, data may be preprocessed by an imaging device before being transmitted to a controller for image detection, or may be preprocessed by a first system (e.g., a controller, a computer, a server, a GPU, etc.) prior to being received by a second system (e.g., controller **400** and/or object detector **420**) for object (e.g., refuse can) detection.

[0067] In some embodiments, preprocessing the data may include any number of functions based on a particular implementation. For example, preprocessing for a one-stage object detector such as object detector **420** may include determining and/or modifying the aspect ratio and/or scaling of received image data, determining or calculating the mean and/or standard deviation of the image data, normalizing the image data, reducing dimensionality (e.g., converting to grey-scale) of the image data, etc. In some embodiments, preprocessing may include determining and/or modifying the image data to ensure that the image data has appropriate object segmentation for utilizing during training (e.g., of object detector **420**) and/or object detection. In some embodiments, preprocessing may include extracting or determining particular frames of video for further processing.

[0068] At step **706**, the data is input into an object detector, such as object detector **420** as described above. The object detector may process the data to detect one or more target objects (e.g., refuse can and/or human beings). Generally, the object detector processes the data as described above with respect to FIGS. 5 and 6. In this manner, the data may be processed by a single-stage object detector (e.g., object detector **420**), such as RetinaNet. The output of the object detector may be an indication of target objects, such as one or more refuse cans, and an indication of a confidence level for the detected objects. As an example, the indication of the target objects may include a class of the object (e.g., “refuse can,” “person,” etc.) and a confidence level that a bounding box (e.g., shown in FIG. 8A) associated with the detected object actually contains the object.

[0069] At step **708**, a determination is made based on the identification of human beings during object detection in the previous step. In some embodiments, the determination is made if a human being is detected within a predefined danger zone (e.g., an area of the image captured by the object sensors). The danger zone may indicate a region (e.g., in the proximity of refuse vehicle **10**) where

a person may be injured if automated refuse collection operations are initiated. If a human being is detected, process **700** continues to step **710**. At step **710**, safety measures may be initiated to prevent harm and/or injury to the person detected in the danger zone. The safety measures may include restricting movement of a refuse vehicle and/or an actuator assembly, such that the vehicle and/or the actuator assembly cannot move to engage a refuse can if a human being is detected within a danger zone. In some embodiments, the safety measures may include presenting an alarm (i.e., a notification) to an operator of the refuse vehicle (e.g., via user interface **432**), to alert the operator to the detected human being.

[0070] If a human being is not detected, process **700** continues to step **712**. At **712**, a determination is made based on the whether or not a refuse can (e.g., or multiple refuse cans) are detected based on the data. In some embodiments, the determination is based on the confidence level associated with a detected object (e.g., associated with a bounding box for the detected object, as shown in FIG. **8A**). In some embodiments, a confidence level at or above a threshold value may indicate a determination that an object (e.g., a refuse can) is detected. A confidence level below the threshold value may indicate a determination that a refuse can is not detected. If no refuse can is detected, process **700** may continue back to step **702**, where the process of capturing and processing data is repeated. If a refuse can is detected, however, process **700** may continue to step **714**.

[0071] At step **714**, a response is initiated based on the detection of a refuse can. The response may include any number of automated control actions. For example, the response may include presenting a notification or indication of the detected refuse can to an operator via a user interface (e.g., user interface **432**). In this example, the operator may be provided with means for selecting one of one or more detected refuse cans to act on (e.g., to move to and engage). As another example, the control actions may include automatically moving the refuse vehicle and/or an actuator assembly to engage the refuse can. The control actions initiated at step **714** are described in detail below, with respect to FIG. **9**.

[0072] Referring now to FIG. **8A**, an example interface **800** illustrating the detection of multiple refuse cans is shown, according to some embodiments. Interface **800** may illustrate an example of a user interface presented to a user of controller **400** and/or refuse vehicle **10**. Interface **800** may be presented via user interface **432**, for example. In some embodiments, interface **800** illustrates the detection of refuse can objects from data captured by one or more image and/or object sensors. In some embodiments, interface **800** may be an example of an interface presented based on process **700**.

[0073] In some embodiments, the image of interface **800** may represent an input image to object detector **420**. Object detector **420** may be configured to detect any number of object classes, as described above, including at least refuse cans. As shown, a first refuse can **802** and a second refuse can **804** have been detected (e.g., by object detector **420**). Each of refuse cans **802** and **804** are shown with a corresponding bounding box, indicating the object within interface **800** and a probability that the bounding box actually contains the detected object. The bounding boxes for each of refuse cans **802** and **804** may not only indicate detected objects, but may indicate a location of each of refuse cans **802** and **804** within a captured images (e.g., the image presented in interface **800**).

[0074] Each of refuse cans **802** and **804** are shown with a corresponding confidence value (e.g., 0.999 and 0.990, respectively). The confidence values may indicate a level of confidence that the associated bounding box actually contains an object (e.g., a refuse can). As described above, objects with a confidence value below a threshold may be ignored (e.g., not presented with a bounding box as shown). In some embodiments, an operator (e.g., of refuse vehicle **10**) may select a refuse can to engage with (e.g., move to, pickup, and empty) from interface **800**. For example, the user may select one of refuse cans **802** or **804** via a user input device (e.g., by touching a particular refuse can via a touchscreen).

[0075] In some embodiments, interface **800** may include a graphic element such as a start button

808 that the user may select to initiate retrieval of the selected refuse can. In other embodiments, retrieval of the selected refuse can may be initiated by selecting a graphical element representing the refuse can (e.g., the image of bounding box of one of refuse cans **802** or **804**). It will be appreciated that interface **800** may include any number of additional graphical elements to facilitate the selection and retrieval of a refuse can. For example, interface **800** may include additional buttons, menus, icons, image, etc.

[0076] Referring now to FIG. **8B**, another example interface **810** illustrating the detection of multiple refuse cans is shown, according to some embodiments. Like interface **800**, interface **810** may illustrate an example of a user interface presented to a user of controller **400** and/or refuse vehicle **10**. Interface **810** may be presented via user interface **432**, for example. In some embodiments, interface **810** represents an alternative configuration of interface **800**.

[0077] As shown, interface **810** includes a top-down view of a path being traversed by refuse vehicle **10**. In this example, interface **810** presents a graphical representation of a roadway. In some embodiments, interface **810** may not include an illustration of the path and may only indicate a position of a refuse can with respect to refuse vehicle **10**. Also shown are multiple graphical elements representing refuse cans, shown as refuse cans **802** and **804** on a left (i.e., passenger) side of the roadway and as refuse can **806** on a right (i.e., driver's) side of the roadway. In this regards, interface **810** illustrates the detection of refuse cans from multiple sides of refuse vehicle **10**.

[0078] In some embodiments, interface **810** is generated from aerial or satellite images of a location of refuse vehicle **10**. For example, satellite imagery may be retrieved via network **440** based on a determined location of refuse vehicle **10**. In this example, the location of refuse vehicle **10** may be determined based on GPS coordinates, triangulation (e.g., via a cellular network), or by any other methods for determining a location. In other embodiments, interface **810** may be generated from images captured by image/object sensors **430** located at various points around refuse vehicle **10**. In such embodiments, multiple images or data may be combined from image/object sensors **430** to form a panoramic or top-down view of the area around refuse vehicle **10**. In yet other embodiments, the background (e.g., the roadway) of interface **810** may be a generated graphical element.

[0079] As described with respect to interface **800**, an operator (e.g., of refuse vehicle **10**) may select a refuse can to engage with (e.g., move to, pickup, and empty) from interface **810**. For example, the user may select one of refuse cans **802**, **804**, or **806** via a user input device (e.g., by touching a particular refuse can via a touchscreen). In some embodiments, the user may select start button **808** to initiate retrieval of the selected refuse can. In other embodiments, retrieval of the selected refuse can may be initiated by selecting a graphical element representing the refuse can (e.g., one of refuse cans **802**, **804**, or **806**). It will be appreciated that interface **810** may include any number of additional graphical elements to facilitate the selection and retrieval of a refuse can. For example, interface **800** may include additional buttons, menus, icons, image, etc.

[0080] Referring now to FIG. **9**, a process **900** for initiating control actions based on a detected refuse can is shown, according to some embodiments. Process **900** may be implemented in response to detecting a refuse can, as described above with respect to process **500**. Similar to process **500**, process **900** may be implemented by a controller of a refuse vehicle (e.g., refuse vehicle **10**), such as controller **400**, described above. In various embodiments, process **900** is a continuation of process **500**, or is performed subsequent to process **500**. As denoted herein, an actuator assembly may refer to any type of grabber and/or lift assembly configured to engage and empty a refuse can into a refuse can of a refuse vehicle. For example, the actuator assembly may be lift assembly **100** or lift assembly **200**, as described above.

[0081] At step **902**, a particular refuse can is identified. As described above, multiple objects including multiple refuse cans may be detected. In order to initiate a control action, a particular refuse can may be identified, either automatically or based on a user input. In the first case, where a particular refuse can is automatically identified in order to initiate a control action, a controller

(e.g., controller **400**) may implement a number of parameters for identifying the particular refuse can. For example, the refuse can may be identified based on identifying features (e.g., size, color, shape, logos or markings, etc.) or may be selected based on its proximity to the refuse vehicle (e.g., the closest refuse can may be identified first). The particular refuse can may be automatically identified in autonomous operations (e.g., where refuse vehicle **10** is autonomous) in order to reduce or eliminate operator input.

[0082] In some embodiments (e.g., semi-autonomous or non-autonomous operations), the particular refuse can may be selected by an operator. As described above, for example, the operator may be presented with a user interface (e.g., interface **800**) for viewing captured data (e.g., image data) and identified objects. The operator may select, from the user interface, the particular refuse can. Using interface **800** as an example, the operator may select one of refuse cans **802** or **804**, in order to initiate collection of the particular refuse can.

[0083] At step **904**, a location of the identified refuse can is determined. In some embodiments, the location of the refuse can may be determined based on the location of the refuse vehicle, such that the location of the refuse can is determined relative to the refuse vehicle. In some embodiments, sensor data from image/object sensors **430** may be used to determine the location of the detected refuse can. For example, data from LIDAR or radar sensors may be used to determine a location of the refuse can, and/or may be used to supplement other data (e.g., from a visible light camera). The determination of a location of a detected refuse can is described in further detail below, with respect to FIG. **10A**.

[0084] At step **906**, a trajectory is generated for the refuse vehicle based on the location of the refuse can. The trajectory of the refuse vehicle, for example, may indicate a path or a set of movements for the refuse vehicle to follow to move to align with the refuse can such that the actuator assembly may move to engage the refuse can. For refuse vehicles without a grabber assembly, for example, the trajectory may indicate a path or movements to align the refuse vehicle with the refuse can (e.g., head-on) such as to engage the refuse can with a fork assembly. For refuse vehicles with a grabber assembly, the trajectory may indicate a path or movements to move the refuse vehicle alongside the refuse can (e.g., to engage the refuse can with the grabber assembly). The generation of a trajectory of the refuse vehicle is described in further detail below, with respect to FIG. **10B**.

[0085] At step **908**, a trajectory is generated for an actuator assembly of the refuse vehicle. In some embodiments, step **908** occurs simultaneously with step **906**. In other embodiments, step **908** occurs prior or subsequently to step **906**. The trajectory of the actuator assembly may indicate a path or a set of movements that the actuator assembly may follow to engage the refuse can once the refuse vehicle has moved alongside the refuse can. In some embodiments, such as with a side loading refuse vehicle, or a refuse vehicle with a grabber assembly, the trajectory may indicate a series of movements lateral, longitudinal, and/or vertical movements that the grabber assembly may follow to retrieve a refuse can. In other embodiments, such as refuse vehicles without a grabber assembly, the trajectory may indicate only longitudinal and/or vertical movements that the grabber assembly may follow to retrieve a refuse can.

[0086] At steps **910** and **912**, the refuse vehicle and actuator assembly navigate (i.e., move) to the refuse can. As with steps **906** and **908**, steps **910** and **912** may occur simultaneous or concurrently. In autonomous and/or semi-autonomous operations, the refuse vehicle (e.g., refuse vehicle **10**) and actuator assembly (e.g., actuator assembly **436**) may be controlled or commanded (e.g., by control module **424**) to automatically navigate to the refuse can. For example, the refuse vehicle may automatically move to the refuse can, and the actuator may automatically move to engage the refuse can, without operator input. In other embodiments, the trajectories generated at steps **906** and **908** may be presented to the operator (e.g., via a user interface) so that the operator may navigate the refuse vehicle and/or the actuator to the refuse can. As an example, the trajectories may be presented via a user interface, indicating a path and/or movements that the operator should

follow to navigate to the refuse can, as shown in FIGS. **11A-11D** below.

[0087] In some embodiments, as the refuse vehicle and/or the actuator assembly navigate (i.e., move) towards the refuse can, image data and/or sensor data may be captured from the various subsystems of the refuse vehicle (e.g., vehicle systems **434**) and/or from the actuator assembly (e.g., actuator assembly **436**). The captured image and/or sensor data may be transmitted to feedback module **426** in order to improve, modify, and/or otherwise adjust the movements of the refuse vehicle and/or actuator assembly. As described above, feedback module **426** may include a RNN for processing feedback data. As an example, feedback module **426** may interpret feedback data on the movement of the actuator assembly to adjust the trajectory of the actuator assembly as it moves to engage the refuse can. In another example, a proposed trajectory presented to an operator may be continuously updated to reflect a current position of the refuse vehicle with respect to the refuse can, as the refuse vehicle moves.

[0088] At step **914**, the refuse can is engaged by the actuator assembly. The refuse can may be engaged by moving the actuator assembly in any suitable direction to engage and lift the refuse can. In some embodiments, such as with a refuse vehicle with a grabber assembly, the actuator assembly may move laterally, longitudinally, and/or vertically to engage the refuse can. In other embodiments, such as refuse vehicles without a grabber assembly, the actuator assembly may only move longitudinally and/or vertically to engage the refuse can. For Once the actuator assembly has secured the refuse can (e.g., by closing actuators, by inserting a fork assembly, etc.), the actuator assembly may lift the refuse can to empty the contents of the refuse can into a refuse compartment (e.g., refuse compartment **30**).

[0089] Referring now to FIG. **10A**, an example diagram illustrating the determination of a location of a refuse can is shown, according to some embodiments. As described above with respect to step **904** of process **900**, for example, a location or position of a refuse can may be determined after the refuse can is detected (e.g., by object detector **420**). Also as described above, data from image/object detectors **430** may be used to determine the location of the refuse can with respect to refuse vehicle **10**. In some embodiments, any other suitable devices or sensors (e.g., GPS sensors, LIDAR, etc.) may be used to determine a position or location of a target refuse can.

[0090] The diagram of FIG. **10A** is shown to include a first point **1002** that represents a location (i.e., position) of a refuse vehicle and a second point **1004** that represents a location of the detected refuse can with respect to the refuse vehicle. A dotted (i.e., broken) line is shown between points **1002** and **1004** which indicates a distance (i.e., magnitude) and a direction between points **1002** and **1004** (e.g., the refuse vehicle and the target refuse can). This distance and direction can be represented by a latitudinal (i.e., vertical) and a longitudinal (i.e., horizontal) component, shown as lines x and y. In some embodiments, the x and y components represent a distance the refuse vehicle must move in a corresponding direction or along a corresponding axis to reach the refuse can. Based on the position of the refuse can with respect to the refuse vehicle, a trajectory can be determined to provide means for the refuse vehicle to move to retrieve/engage the refuse can.

[0091] Referring now to FIG. **10B**, an example diagram illustrating the determination of a trajectory of refuse vehicle **10** based on the determined location of a detected refuse can **1012** is shown, according to some embodiments. More specifically, in the example diagram of FIG. **10B**, a trajectory is determined for a front loading configuration of refuse vehicle **10** that is retrieving refuse can **1012** which may be an industrial or commercial refuse can configured to be engaged with a fork assembly. It will be appreciated, however, that similar methods to those described herein may be used to determine a trajectory for any configuration of refuse vehicle **10** (e.g., front loading, side loading, rear loading) to retrieve any type of refuse can.

[0092] As shown in FIG. **10B**, a current trajectory of refuse vehicle **10** is represented by a first arrow a. The current trajectory of refuse vehicle **10** may represent a trajectory that the vehicle would follow if it continued on a current path (i.e., moved forward or longitudinally). A projected trajectory is shown by a second arrow b. The proposed trajectory may indicate a trajectory that

refuse vehicle **10** may follow to reach a second position where refuse vehicle **10** is aligned with refuse can **1012**. The second position is indicated with dotted (i.e., broken) lines. In the second position, as shown in FIG. **10B**, refuse vehicle **10** may be aligned with refuse can **1012** such that refuse vehicle **10** may drive substantially forward to engage refuse can **1012**. In another example, where refuse truck **10** is in a side loading configuration, the second position may be orthogonal or alongside of refuse can **1012** to engage refuse can **1012** from a side of refuse vehicle **10**.

[0093] In some embodiments, the proposed trajectory is defined by a distance (i.e., magnitude) and a yaw (i.e., angle about a normal axis) that refuse vehicle **10** may follow to reach refuse can **1012**. As shown, for example, the yaw is represented by an angle ψ which indicates a number of degrees that refuse vehicle **10** must turn left or right (e.g., with respect to the current trajectory) to reach the second position. In some embodiments, angle ψ may be continuously determined as refuse vehicle **10** moves towards the second position, in line with refuse can **1012**. In other words, as refuse vehicle **10** moves toward the refuse can, the proposed trajectory may be continuously determined or updated to reflect a new position of the refuse vehicle.

[0094] In some embodiments, a current position of refuse vehicle **10** is continuously updated or determined such that the proposed trajectory is continuously updated or determined. In such embodiment, any number of sensors or devices may be used to determine the trajectory. For example, the position and movement of refuse vehicle **10** may be determined based on GPS sensors, cameras or object sensors (e.g., image/object sensors **420**), inertial sensors, etc. In some embodiments, the data from any of these sensors is processed by controller **400** (e.g., by feedback module **426**).

[0095] In some embodiments, such as when refuse vehicle **10** is a front loading refuse vehicle, the positioning of refuse vehicle **10** with respect to refuse can **1012** may be particularly important. For example, in some cases, such as with front loading refuse vehicles having fork attachments, an actuator assembly may have a limited range of motion in one or more planes. With a front loading refuse vehicle having a fork attachment, for example, the fork attachment may not be able to move left or right (i.e., laterally). In such embodiments, it may be necessary to align refuse vehicle **10** such that refuse vehicle **10** can drive substantially straight forward to engage refuse can **1012**. This may minimize operator input, removing or reducing the need for the operator to exit refuse vehicle **10** to manually move refuse can **1012** into position in front of refuse vehicle **10**.

[0096] Referring now to FIGS. **11A-11D**, example user interfaces for presenting the determined trajectory for refuse vehicle **10** are shown, according to some embodiments. These example user interfaces may be generated by UI manager **422**, for example, and/or presented via user interface **432** (e.g., to a user of refuse vehicle **10**). In various embodiments, these example user interfaces may be an overlay of another user interface (e.g., interface **800** or **810** described above), may be presented as via a heads-up display (HUD), may occupy at least a majority of a user interface, or may be presented by any other suitable methods.

[0097] It will be appreciated the example interfaces shown are not intended to be limiting and that any suitable interface or graphical elements for presenting similar information may be used. In some embodiments, the example user interfaces of FIGS. **11A-11D** may include a variety of other information based on operator preference, operation requirements, etc., such as menus, other graphical elements, operating information, etc. As described above with respect to interface **800** or **810**, for example, any of the interfaces of FIGS. **11A-11D** may include generated images, aerial or satellite imagery, or any other suitable method for generating a user interface.

[0098] The example user interfaces shown in FIGS. **11A-11D** may be presented to an operator of refuse vehicle **10** to aid the operator in navigating refuse vehicle **10** to a refuse can, such as in semi-autonomous or manual configurations, and/or may illustrate a path taken by refuse vehicle **10** in autonomous configurations (e.g., without operator input). As an example, an example user interface such as the interface shown in FIG. **11A** may be presented to guide the user in navigating refuse vehicle **10**. In this example, the user may operate refuse vehicle **10** by following the path illustrated

on the interface. In semi-autonomous operations, for example, such a user interface may allow the user to view a projected path (i.e., trajectory) before the user determines whether or not refuse vehicle **10** should move along the path autonomously.

[0099] In some embodiments, a projected path may be shown as a graphical element **1106**. As described above with respect to FIGS. **10A** and **10B**, the projected path is based on the determined trajectory and may be continuously updated or determined as refuse vehicle **10** moves. As shown in FIGS. **11A** and **11B**, for example, graphical element **1106** may be an arrow, a line, or any other suitable graphical element for illustrating a proposed path (i.e., trajectory) that refuse vehicle **10** may take to reach a target refuse can. In this example, the target refuse can is shown as a graphical element **1104**, while refuse vehicle **10** is shown as a graphical element **1102**. FIG. **11A**, in particular, illustrates refuse vehicle **10** traveling down a path (e.g., a two-lane roadway) to engage a refuse can (e.g., graphical element **1104**). In this example, graphical element **1106** shows that refuse vehicle **10** may drive forward, turn slightly right to move toward a curb or an edge of the path, and subsequently turn slightly left to a position orthogonal to the refuse can, in order to engage the refuse can via a grabber assembly mounted on the side of the refuse vehicle or attached to an intermediate storage container.

[0100] FIG. **11B** illustrates a top-down view of refuse vehicle **10**, shown again as graphical element **1102**, which may follow the path illustrated by graphical element **1106** to engage a front loading refuse can, shown as graphical element **1104**. In this example, refuse vehicle **10** may turn slightly left while driving towards the target refuse can and subsequently turn slight right to align with the refuse can. In this example, the proposed path may align refuse vehicle **10** with the refuse can such that refuse vehicle **10** may drive substantially straight forward to engage the refuse can. As described above, aligning refuse vehicle **10** with a target refuse can may be particularly critical to reduce or eliminate manual intervention from an operator.

[0101] In some embodiments, a proposed path or trajectory may be shown in another manner. As shown in FIG. **11C**, for example, the proposed trajectory may be illustrated by an arrow, shown as graphical element **1108**, or other similar graphical element that indicates a direction of travel to an operator of refuse vehicle **10**. Graphical element **1108** may be presented as a HUD, for example, where the interface of FIG. **11C** illustrates a view of the operator of refuse vehicle **10**. In another example, the interface of FIG. **11C** may illustrate a virtual reality or augmented reality view from the front of refuse vehicle **10**.

[0102] As shown in FIG. **11D**, a proposed path or trajectory, shown as graphical element **1110**, may be presented in a perspective view. As with the interface of FIG. **11C**, for example, the interface of FIG. **11D** may represent a virtual reality or augmented reality view from the front of refuse vehicle **10**. In various embodiments, graphical element **1110** may be presented via a HUD or in another type of user interface. Graphical element **1110** illustrates a project path for refuse vehicle **10** to move forward to engage a front loading refuse can. In this regard, the path illustrated by graphical element **1110** may align the front of refuse vehicle **10** with the refuse can so that the refuse can may be engaged by a fork assembly.

[0103] As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

[0104] The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or

movable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

[0105] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0106] Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

[0107] The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

[0108] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products including machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Claims

1. A system, comprising: a sensor configured to detect objects on one or more sides of a refuse collection vehicle; an actuator assembly configured to actuate to engage a refuse can; and a controller configured to: detect, using a single-stage object detector, a presence of the refuse can; determine, based on data obtained by the sensor, a position of the refuse can with respect to the refuse collection vehicle; generate a trajectory that indicates a path of at least one of the actuator assembly or the refuse collection vehicle and that indicates an alignment between (i) the at least one of the actuator assembly or the refuse collection vehicle, and (ii) the refuse can; and present, via a user interface, the trajectory.
2. The system of claim 1, wherein the trajectory includes one or more first characteristics to indicate the alignment or includes one or more second characteristics to indicate unalignment.
3. The system of claim 1, wherein the trajectory includes a line that extends from the at least one of the actuator assembly or the refuse collection vehicle and towards the refuse can to indicate the alignment.
4. The system of claim 1, wherein the controller is further configured to: receive, via the user interface, a selection to accept the trajectory; and initiate a control action to cause the refuse collection vehicle to perform at least one operation to align the refuse collection vehicle with the refuse can.
5. The system of claim 1, further comprising: the user interface to include: a graphical representation of at least a portion of the refuse collection vehicle; and a graphical representation of the refuse can.
6. The system of claim 1, wherein the controller is further configured to: detect, based on an output of the single-stage object detector, a second refuse can; determine a position of the second refuse can; update the user interface to include (i) a first graphical representation of the refuse can and (ii) a second graphical representation of the second refuse can; and receive, responsive to an interaction with the first graphical representation, a selection to accept the trajectory.
7. The system of claim 1, wherein the controller is further configured to: present the user interface to include a graphical representation of the refuse can; detect, via the user interface, a selection of the graphical representation of the refuse can; update, responsive to detection of the selection of the graphical representation of the refuse can, the user interface to include an element to confirm selection of the refuse can; and receive, responsive to an interaction with the element, confirmation of the selection of the graphical representation.
8. The system of claim 1, wherein the sensor includes at least one camera, and wherein the controller is configured to: receive, from the at least one camera, at least one image frame previously provided to the single-stage object detector; and present the user interface to further include an indication of the position of the refuse can within the at least one image frame.
9. The system of claim 1, wherein the controller is further configured to: detect, based on an output of the single-stage object detector, a classification of the refuse can; and update, responsive to detection of the classification, the user interface to include an indication of the classification.
10. The system of claim 1, wherein the one or more sides include a first side and a second side opposite the first side, and wherein the controller is further configured to: detect the presence of the refuse can on the first side; detect a presence of a second refuse can on the second side; and present the user interface to include (i) a graphical representation of the refuse can and (ii) a graphical representation of the second refuse can.
11. The system of claim 1, wherein the controller is further configured to: monitor subsequent movement of the refuse collection vehicle with respect to the trajectory; generate, based on the subsequent movement of the refuse collection vehicle, an updated version of the trajectory; and update the user interface to present the updated version of the trajectory.

- 12.** The system of claim 1, where the controller is further configured to: detect, based on second data collected by the sensor, a traversal of the refuse collection vehicle along the path; and update the user interface to reflect the traversal of the refuse collection vehicle along the path.
- 13.** The system of claim 1, wherein the controller is further configured to: cause a heads-up-display (HUD) of the refuse collection vehicle to present the user interface.
- 14.** An object detection system for inclusion in a refuse collection vehicle, the object detection system comprising: one or more processing circuits configured to: detect, based at least on an output of a single-stage object detector, a presence of a refuse can; determine, based on data obtained by at least one sensor disposed on the refuse collection vehicle, a position of the refuse can with respect to the refuse collection vehicle; generate a trajectory that indicates a path of at least one of an actuator assembly of the refuse collection vehicle or the refuse collection vehicle and that indicates an alignment between (i) the at least one of the actuator assembly or the refuse collection vehicle, and (ii) the refuse can; and present, via a user interface, the trajectory.
- 15.** The object detection system of claim 14, wherein the one or more processing circuits are further configured to: detect, based on a second output of the single-stage object detector, a second refuse can; determine a position of the second refuse can; and update the user interface to include a graphical representation of the second refuse can.
- 16.** The object detection system of claim 14, wherein the at least one sensor includes at least one camera, and wherein the one or more processing circuits are further configured to: receive, from the at least one camera, at least one image frame previously provided to the single-stage object detector; and present the user interface to further include an indication of the position of the refuse can within the at least one image frame.
- 17.** The object detection system of claim 14, wherein the refuse collection vehicle includes a first side and a second side opposite the first side, and wherein the one or more processing circuits are further configured to: detect the presence of the refuse can on the first side; detect a presence of a second refuse can on the second side; and present the user interface to include (i) a first graphical representation of the refuse can and (ii) a second graphical representation of the second refuse can.
- 18.** The object detection system of claim 14, wherein the one or more processing circuits are further configured to: monitor subsequent movement of the refuse collection vehicle with respect to the trajectory; generate, based on the subsequent movement of the refuse collection vehicle, an updated version of the trajectory; and update the user interface to present the updated version of the trajectory.
- 19.** A refuse collection vehicle, comprising: an assembly configured to actuate to engage a refuse can; a sensor disposed on an external surface of the assembly; and one or more processing circuits configured to: detect, based at least on an output of a single-stage object detector, a presence of a refuse can; determine, based on data obtained by the sensor, a position of the refuse can with respect to the refuse collection vehicle; generate a trajectory that indicates a path of at least one of the assembly or the refuse collection vehicle and that indicates an alignment between (i) the at least one of the assembly or the refuse collection vehicle, and (ii) the refuse can; and present, via a user interface, the trajectory.
- 20.** The refuse collection vehicle of claim 19, wherein the trajectory further includes at least one of: one or more first characteristics to indicate the alignment; one or more second characteristics to indicate unalignment; or a line that extends from the at least one of the assembly or the refuse collection vehicle and towards the refuse can to indicate the alignment.
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